

RAJALAKSHMI ENGINEERING COLLEGE

Experiment 2: EDA - Data Import and Export

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AIM

To create pandas Series and DataFrames, import data from Excel, web and SQLite sources, and export the processed data to Excel.

SOURCE

EXPT2/FDVA2.ipynb

PYTHON CODE - CELL 1

```
import pandas as pd
import sqlite3
```

PYTHON CODE - CELL 2

```
data_dict = {
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [25, 30, 35],
    "City": ["Chennai", "Mumbai", "Delhi"]
}
print("Dictionary:")
print(data_dict)
```

EXECUTED OUTPUT

```
Dictionary:
{'Name': ['Alice', 'Bob', 'Charlie'], 'Age': [25, 30, 35], 'City': ['Chennai', 'Mumbai', 'Delhi']}
```

PYTHON CODE - CELL 3

```
# 3. Create Series
# =====
series_data = pd.Series([10, 20, 30, 40])
print("\nSeries:")
print(series_data)
```

EXECUTED OUTPUT

```
Series:
0    10
1    20
2    30
3    40
dtype: int64
```

PYTHON CODE - CELL 4

```
# 4. Create DataFrame
# =====
df = pd.DataFrame(data_dict)
print("\nDataFrame:")
print(df)
```

EXECUTED OUTPUT

```
DataFrame:
   Name  Age  City
0  Alice   25  Chennai
1   Bob   30  Mumbai
2  Charlie  35   Delhi
```

PYTHON CODE - CELL 5

```
df = pd.read_excel("sample_data.xlsx")
df.head()
```

EXECUTED OUTPUT

	Name	Age	City	Salary
0	Alice	25	Chennai	30000
1	Bob	30	Mumbai	40000
2	Charlie	35	Delhi	50000
3	David	28	Bangalore	35000
4	Eva	26	Hyderabad	32000

PYTHON CODE - CELL 6

```
import pandas as pd
import requests
from io import StringIO
url = "https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)"

headers = {
    "User-Agent": "Mozilla/5.0"
}

response = requests.get(url, headers=headers)

# Wrap HTML with StringIO (fix warning)
tables = pd.read_html(StringIO(response.text))

df_web = tables[0]
print(df_web.head())
```

EXECUTED OUTPUT

```
0 Largest economies in the world by nominal GDP ...
```

PYTHON CODE - CELL 7

```
conn = sqlite3.connect("sample.db")

# Save DataFrame to SQL
df.to_sql("people", conn, if_exists="replace", index=False)

# Read from SQL
df_sql = pd.read_sql("SELECT * FROM people", conn)
print("\nSQL Data:")
print(df_sql)

conn.close()
```

EXECUTED OUTPUT

```
SQL Data:
   Name  Age  City  Salary
0  Alice   25  Chennai  30000
1    Bob   30  Mumbai  40000
2  Charlie  35   Delhi  50000
3   David  28  Bangalore  35000
4     Eva  26  Hyderabad  32000
```

PYTHON CODE - CELL 8

```
# 9. Export to Excel
# =====
df_sql.to_excel("output.xlsx", index=False)

print("\nExcel file 'output.xlsx' created successfully!")
```

EXECUTED OUTPUT

```
Excel file 'output.xlsx' created successfully!
```

RESULT

Data was created, imported from the available sources, stored and retrieved through SQLite, and exported to Excel successfully, as shown by the executed outputs.